PROJECT CHARTER

Synergy-Aware Assist Controller

A Simulation-Based Model of Shoulder-Abduction-Dependent Elbow-Flexion Compensation During Post-Stroke Reaching

Independent Portfolio Project
Rehabilitation Robotics, Computational Modelling & Control

Version 3.0 - Revised Technical Charter

Executive Summary

This project develops and evaluates a simulation-based rehabilitation-robotics controller targeting a clinically documented post-stroke motor impairment: abnormal coupling between shoulder-abduction effort and elbow flexion during reaching.

Following stroke, particularly in individuals with moderate-to-severe hemiparesis, increasing shoulder-abduction demand can increase involuntary elbow-flexion expression and restrict reaching workspace. This phenomenon has been characterised experimentally using robotic paradigms in which shoulder-abduction loading is systematically manipulated.

Primary question: Can a model-based, synergy-aware assistive controller detect and compensate for simulated shoulder-abduction-dependent elbow-flexion disturbances during reaching, while preserving task performance and minimizing robotic assistance?

The project does not attempt to reproduce the full neurophysiology of post-stroke flexor synergy, demonstrate clinical efficacy, or establish motor recovery.

Instead, it develops a deliberately reduced-order 2.5D simulation consisting of:

1. a two-degree-of-freedom planar upper-limb reaching model;
2. an independent scalar representing shoulder-abduction demand;
3. a parameterised phenomenological impairment model;
4. a simulated voluntary motor command;
5. a conventional impedance-based robotic controller;
6. an online synergy estimator using observable task and kinematic information;
7. a constrained synergy-aware corrective controller; and
8. an experimental framework comparing unassisted, conventional assistance, generic additional assistance and synergy-aware assistance.

The central methodological principle is separation between simulated human motor output, impairment, arm behaviour, detection, and robotic compensation. The impairment therefore acts on simulated human motor output rather than corrupting the robot controller's own command.

The project will first verify that the simulator implements its equations correctly, then evaluate whether the impairment model demonstrates literature-consistent phenomenological behaviour, before evaluating the proposed controller.

Success means demonstrating that, within the validated synthetic model, synergy-aware assistance reduces abduction-dependent unwanted elbow flexion while maintaining reaching performance and smoothness and avoiding unnecessary robotic assistance. It does not imply that pathological synergy has been neurologically reduced or that motor recovery would occur in a person with stroke.

1. Project Overview

The Synergy-Aware Assist Controller is a standalone computational rehabilitation-robotics project investigating movement-quality-aware robotic assistance.

Upper-limb rehabilitation robots commonly use trajectory tracking, impedance control, admittance control, assist-as-needed control, or related approaches to facilitate movement. However, successful endpoint performance does not necessarily imply physiologically desirable joint coordination.

A simulated patient could reach the target while displaying abnormal shoulder-elbow coordination, and a sufficiently strong robot could make an impaired trajectory appear normal without altering the underlying impairment.

1.1 Task performance

Did the simulated hand reach its intended target accurately and within an acceptable duration?

1.2 Movement quality

Did the simulated reach exhibit excessive elbow flexion associated with increasing shoulder-abduction demand?

1.3 Robotic compensation

How much robotic intervention was required to achieve the observed movement?

The project evaluates whether a controller explicitly informed by estimated abnormal coupling can improve movement quality without obtaining that improvement merely by applying substantially greater robotic assistance. The intended output is a reproducible computational demonstration suitable for a robotics/neurorehabilitation portfolio and as groundwork for possible future research using experimental data.

2. Problem Statement

2.1 Clinical problem

Following stroke, loss of independent joint control can contribute substantially to upper-limb reaching impairment. One documented pattern is the expression of abnormal shoulder-abduction/elbow-flexion coupling. Increasing shoulder-abduction demand can increase elbow-flexion tendency and reduce reaching workspace in the paretic arm.

This phenomenon should not be reduced conceptually to simple muscle weakness. It concerns impaired independent control of joint torques and movements and has been investigated using robotic paradigms that manipulate shoulder-abduction loading.

The present project models the observable mechanical consequence of this phenomenon. It does not model its complete neural origin. Although altered corticospinal and brainstem-mediated motor pathways have been proposed as mechanisms contributing to post-stroke flexion synergy, the present simulator will not contain explicit neural pathways, motor-unit recruitment or muscle physiology.

2.2 Engineering problem

A rehabilitation robot can improve endpoint tracking without necessarily improving the underlying quality of simulated joint coordination.

$$\text{successful reach} \;!=\; \text{independent joint control}$$

A controller could also trivially suppress the simulated impairment by imposing large elbow-extension torques. Such a result would have little value unless the amount of robotic assistance, stability and task performance are simultaneously evaluated.

Engineering question: Can the robot detect an abnormal, load-dependent component of the movement and selectively compensate for it while maintaining task performance and limiting unnecessary assistance?

3. Research Questions

Primary Research Question

Can a model-based synergy-aware controller detect and compensate for simulated shoulder-abduction-dependent elbow-flexion disturbances during planar reaching while preserving task performance and minimizing robotic assistance?

Secondary Research Questions

- RQ1 - Impairment validity: Does the phenomenological impairment model reproduce the expected qualitative relationship between increasing shoulder-abduction demand and increasing unwanted elbow-flexion behaviour?
- RQ2 - Detection: Can an online estimator distinguish simulated abnormal coupling from normal task-dependent shoulder-elbow coordination?
- RQ3 - Control: Does synergy-aware assistance reduce the targeted elbow-flexion disturbance more effectively than conventional impedance assistance?
- RQ4 - Specificity: Does synergy-aware assistance outperform generic additional elbow assistance when the assistance budget is controlled?
- RQ5 - Robustness: Does the controller retain acceptable performance across impairment severities, shoulder-abduction demands, reaching directions, model parameters, and simulated sensor noise?
- RQ6 - Efficiency: Can improved movement quality be achieved without excessive robot torque or substantial degradation of movement smoothness and task performance?

4. Objectives

- **O1 -** Develop and verify a two-link planar upper-limb model representing shoulder and elbow motion during reaching.
- **O2 -** Generate an unimpaired reaching command capable of producing physiologically plausible minimum-jerk-like endpoint trajectories.
- **O3 -** Represent shoulder-abduction demand using a separate normalized scalar u_abd in [0,1], rather than treating shoulder-abduction loading as an in-plane joint angle.
- **O4 -** Develop a parameterised phenomenological model in which shoulder-abduction demand produces an additional elbow-flexion torque whose magnitude depends on impairment severity.
- **O5 -** Separate numerical verification, internal model-consistency testing, and literature-constrained phenomenological validation.
- **O6 -** Develop an impedance-based baseline controller against which the proposed controller can be compared.
- **O7 -** Estimate abnormal load-dependent elbow-flexion behaviour using quantities available to a realistic controller rather than privileged simulator ground truth.
- **O8 -** Apply bounded corrective assistance in response to the estimated disturbance.
- **O9 -** Compare impaired/unassisted, conventional impedance assistance, generic additional elbow assistance, and synergy-aware assistance.
- **O10 -** Evaluate the system across multiple impairment levels, shoulder-abduction demands, reaching conditions and model perturbations.
- **O11 -** Publish documented code, experiments, figures and a demonstration suitable for independent reproduction.

5. Scope

5.1 In Scope

- Two-link planar upper-limb dynamics; shoulder and elbow reaching coordinates.
- Independent scalar shoulder-abduction-demand input.
- Parameterised phenomenological impairment model with multiple simulated severities.
- Simulated voluntary reaching and minimum-jerk reference trajectories.
- Conventional impedance/PD-type assistance.

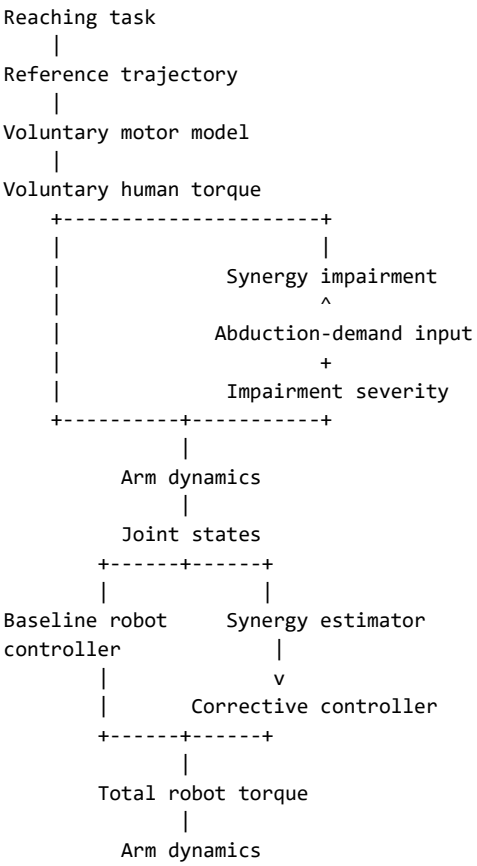
- Online synergy estimation and synergy-aware corrective assistance.
- Torque and torque-rate constraints.
- Multiple shoulder-abduction-demand conditions and reaching directions.
- Sensor-noise experiments and parameter perturbation/model mismatch.
- Controller ablations; kinematic, kinetic and controller-effort metrics.
- Python implementation, reproducible experiments, public repository and demonstration video.

5.2 Out of Scope

- Full 3D shoulder biomechanics; detailed scapulothoracic mechanics; musculoskeletal modelling; muscle activation dynamics; spasticity modelling.
- Explicit corticospinal/reticulospinal modelling or EMG generation.
- Real patient data, human-subject experiments, clinical validation, therapeutic-efficacy claims, or claims of neurological recovery.
- Hardware implementation, actuator design, ROS deployment.
- Reinforcement learning or deep learning as required components.
- Long-term motor-learning simulation.

6. System Architecture

The simulator will explicitly separate the simulated participant from the robotic controller. A separate simulator-only ground-truth channel will record the actual imposed impairment torque. This ground truth may be used for evaluation, but must not be supplied to the online detector.



7. Mathematical Model

7.1 Generalised coordinates

$$q = [q_s, q_e]^T$$

q_s is the shoulder angle in the reaching plane and q_e is the elbow angle.

$$u_abd \text{ in } [0, 1]$$

u_abd is normalized shoulder-abduction demand. It is a virtual effort/load input, not an anatomical shoulder-abduction angle.

8. Arm Dynamics

$$M(q)\ q_ddot + C(q,q_dot)\ q_dot + g(q) + B\ q_dot = tau_human + tau_robot$$

- M(q): configuration-dependent inertia matrix.
- C(q,q_dot)q_dot: Coriolis/centrifugal terms.
- g(q): gravitational contribution where applicable.
- B q_dot: viscous damping.
- tau_human: simulated participant-generated torque.
- tau_robot: robot-generated torque.

For a horizontal planar reaching model, gravity acting on the planar coordinates may be neglected. Shoulder-abduction loading remains represented separately through u_abd.

9. Simulated Human Motor Output

$$tau_human = tau_vol + tau_syn$$

tau_vol is intended voluntary motor command and tau_syn is abnormal synthetic coupling torque. This separation is fundamental: the pathological component is simulated as an impairment of human motor output, not a corruption of the robot's control signal.

10. Shoulder-Abduction-Dependent Impairment Model

$$tau_syn,e = alpha\ f(u_abd)\ tau_scale$$

- tau_syn,e: additional elbow-flexion torque.
- alpha: dimensionless impairment-severity parameter.
- u_abd: normalized shoulder-abduction demand.
- f(.): specified load-response function.
- tau_scale: reference torque scaling factor.

The simplest initial hypothesis is f(u_abd) = u_abd, giving:

$$tau_syn,e = alpha\ u_abd\ tau_scale$$

This linear relationship is an engineering abstraction, not a claimed physiological law. Alternative nonlinear functions may be investigated only if justified by experimental literature or required by model-validation results.

Suggested initial severity set: alpha = {0, alpha_mild, alpha_moderate, alpha_severe}. Numerical values will be documented and justified separately.

11. Baseline Robotic Controller

$$\text{tau_base} = \text{K_p}\ (\text{q_d} - \text{q}) + \text{K_d}\ (\text{q_dot_d} - \text{q_dot})$$

q_d is the desired joint trajectory, K_p the proportional/stiffness matrix, and K_d the derivative/damping matrix. This controller serves as the conventional tracking-assistance comparator and will be tuned first under the unimpaired condition. The project will not describe this simplified implementation as reproducing the complete general impedance-control framework.

12. Synergy Estimator

12.1 Problem

Normal reaching inherently requires coordinated shoulder and elbow movement. Raw shoulder-elbow correlation is therefore insufficient for detecting pathological coupling. The estimator must identify excess elbow-flexion behaviour relative to expected task-dependent movement and examine whether that excess is associated with shoulder-abduction demand.

12.2 Candidate residual

$$\text{e_syn}(t) = \text{q_e}(t) - \text{q_e,expected}(t)$$

A causal filtered estimate may be computed over a sliding window:

$$\text{e_hat_syn}(t) = (1/N)\ \text{sum}[k=0..N\text{-}1]\ \text{e_syn}(t\text{-}k)$$

The detector may use measured q, measured or estimated q_dot, task reference, commanded shoulder-abduction demand, and previous robot commands. It must not use the simulator's true tau_syn.

12.3 Detector outputs

- Estimated synergy residual.
- Estimated coupling strength.
- Detection state/confidence.
- Detection latency.

Performance will be evaluated against simulator ground truth.

13. Synergy-Aware Corrective Controller

$$\text{tau_assist,e} = \text{-K_c e_hat_syn} - \text{D_c d(e_hat_syn)/dt}$$

Corrective torque is applied only when the estimated disturbance exceeds a defined threshold.

$$\text{tau_robot} = \text{tau_base} + \text{tau_assist}$$

The controller's purpose is not to force the simulated arm onto the reference trajectory regardless of cost. It should provide the minimum useful correction required to reduce the estimated abnormal component.

14. Controller Constraints

$$|\text{tau_robot,i}| <= \text{tau_i,max}$$

$$|\text{d(tau_robot,i)/dt}| <= \text{r_i,max}$$

- Joint-position limits.
- Joint-velocity limits.

- Bounded detector output.
- Bounded controller gains.
- Fixed simulation timestep.
- Documented controller sampling frequency.
- Numerical-instability detection.
- Fallback to baseline control if the estimator becomes invalid.

Torque saturation alone will not be treated as proof of stability. Closed-loop behaviour will be evaluated numerically throughout the predefined operating region.

15. Experimental Conditions

Factor	Conditions
Impairment severity	None; mild; moderate; severe
Shoulder-abduction demand	Low; medium; high
Reaching task	At least three reaching directions or targets
Controller condition	C0 impaired/unassisted; C1 conventional impedance; C2 generic matched elbow assistance; C3 synergy-aware assistance

Where possible, C2 and C3 will be compared under matched torque/assistance budgets. This tests whether the proposed method provides benefits beyond simply applying more elbow-extension torque.

16. Evaluation Metrics

16.1 Primary: excess elbow-flexion error

$$E_AFE = (1/T)\ integral[0..T]\ \{\ max(0,\ q_e(t)\ -\ q_e,expected(t))\ \}\^2\ dt$$

This quantifies task-relative unwanted elbow flexion for the chosen elbow-angle convention. It is not, by itself, interpreted as a direct physiological measure of pathological synergy.

16.2 Abduction-dependent effect

$$Delta\ E_AFE = E_AFE(u_h)\ -\ E_AFE(u_l)$$

This evaluates whether elbow-flexion error increases with abduction demand for matched reaching tasks.

16.3 Coupling gain

$$G_syn = partial(tau_syn,e)\ /\ partial(tau_abd)$$

This is available as simulation ground truth and is not automatically assumed to be directly measurable in a future physical system.

16.4 Endpoint performance

- Final endpoint error.
- Trajectory error.
- Time to target.
- Reach-completion rate.

16.5 Robot assistance effort

$$J_tau = integral[0..T]\ ||tau_robot(t)||\^2\ dt$$

This prevents a controller from being judged successful merely because it applies substantially more torque.

16.6 Movement smoothness

Use a duration- and amplitude-normalized jerk metric or another explicitly defined smoothness measure. Filtering and numerical-differentiation procedures will be reported. Smoothness remains a secondary metric.

16.7 Detector performance

- Estimation error.
- Detection latency.
- False-positive behaviour.
- False-negative behaviour.
- Performance across load levels.
- Performance under noise.

16.8 Constraint/stability metrics

- Peak robot torque.
- Peak torque rate.
- Saturation frequency.
- Joint-limit violations.
- Numerical instability.
- Oscillatory behaviour.
- Unsuccessful trials.

Failures will be reported rather than removed.

17. Controller Objective

$$J = w_e E_{AFE} + w_p E_{task} + w_u J_{tau}$$

The weights define the engineering trade-off between unwanted elbow-flexion behaviour, task performance and robotic assistance. The project does not require solving this objective through optimal control; it provides a formal framework for evaluating controller tuning.

18. Model Verification and Validation

Level 1 - Numerical verification

- Dimensional consistency.
- Zero-input behaviour.
- Open-loop torque response.
- Sensible energy behaviour where appropriate.
- Timestep convergence.
- Joint-limit behaviour.
- Unit tests for individual equations.
- Comparison with analytically predictable simple cases.

Level 2 - Internal model consistency

- alpha = 0 produces no impairment torque.
- u_abd = 0 produces no load-dependent impairment component under the initial model.

- Increasing alpha increases impairment monotonically.
- Increasing u_abd increases impairment monotonically.
- The imposed torque acts in the defined elbow-flexion direction.

Passing these tests verifies implementation consistency; it does not establish physiological validity.

Level 3 - Literature-constrained phenomenological validation

The principal expected behaviour is that increasing shoulder-abduction demand produces increasing difficulty extending/reaching with the elbow under simulated impairment.

- Direction of load-response effects.
- Change in reachable workspace.
- Increasing elbow-flexion tendency with shoulder-abduction loading.
- Qualitative severity dependence.

Quantitative calibration will only be claimed where published data permit defensible parameter extraction.

Level 4 - Robustness / predictive challenge

- Altered arm parameters.
- Unseen severity values.
- Unseen reaching directions.
- Sensor noise.
- Altered abduction demand.

This tests computational generalization within the simulator and does not constitute clinical external validation.

19. Success Criteria

Thresholds will be established during a calibration phase and frozen before final controller comparisons. Final evaluation combinations will not be used to repeatedly retune those thresholds.

19.1 Model criteria

- Arm dynamics pass verification tests.
- Impairment model passes internal-consistency tests.
- Load-response behaviour shows defensible qualitative correspondence with published experimental findings.
- Model assumptions and unsupported parameters are explicitly documented.

19.2 Detector criteria

- Coupling estimates vary appropriately with simulated impairment/load.
- False detection under the unimpaired condition remains acceptably low.
- Performance remains usable under predefined simulated noise.
- Detector does not access hidden simulator ground truth.

19.3 Controller criteria

- Lower abduction-dependent excess elbow-flexion error than conventional assistance.
- Maintained or improved endpoint performance.
- No material deterioration in smoothness.
- Bounded actuator effort.
- No unacceptable instability.

19.4 Specificity criterion

The proposed controller must demonstrate an advantage over generic additional elbow assistance or explicitly conclude that no synergy-specific advantage was demonstrated. A null result is considered a valid project outcome.

20. Interpretation of Success

Strongest permitted conclusion: Within the specified phenomenological impairment model, the synergy-aware controller reduced simulated shoulder-abduction-dependent elbow-flexion disturbance while preserving task performance under the evaluated conditions.

The project will not conclude that pathological synergy was neurologically reduced, reticulospinal drive was modified, independent motor control was restored in humans, motor learning occurred, stroke recovery improved, or the controller is clinically effective.

21. Motor-Learning Interpretation

The project is motivated by rehabilitation, but Version 1 is fundamentally a robotic compensation study, not a motor-learning experiment. A controller that improves movement during assistance may not improve unassisted capability, and excessive assistance could theoretically reduce active engagement.

- Robotic effort will be measured.
- Unnecessary assistance will be penalized.
- Generic high-gain correction will not automatically count as success.
- Assistance-as-needed principles will inform controller tuning.

Future extensions could investigate adaptation, fading assistance, retention and transfer; these are outside the current scope.

22. Novelty Position

The project does not claim that shoulder-abduction/elbow-flexion coupling is newly discovered, that robotic quantification of flexion synergy is new, that progressive shoulder-abduction loading is new, or that rehabilitation robots have never targeted abnormal coordination.

Potential contribution: An interpretable computational framework combining online estimation of load-dependent abnormal movement with constrained, synergy-aware corrective assistance and matched-assistance evaluation in a reproducible simulation.

Novelty will be treated as provisional pending further literature review.

23. Role of Machine Learning and AI

Machine learning is not required for Version 1. The baseline architecture is deliberately model-based and interpretable. Version 1 will not be described as an AI-driven controller unless an actual ML component is subsequently implemented.

- Future ML could support patient-specific impairment estimation.
- Nonlinear coupling estimation.
- State estimation from noisy sensors.
- Adaptation of controller parameters.
- Prediction of assistance requirements.

Any ML model would be compared against the interpretable baseline rather than added solely for novelty.

24. Deliverables

9. Verified 2-DOF arm simulator.
10. Scalar shoulder-abduction-demand implementation.
11. Simulated voluntary reaching model.
12. Parameterised impairment model.
13. Verification test suite.
14. Literature-constrained model-validation notebook.
15. Conventional impedance controller.
16. Online synergy estimator.
17. Generic-assistance comparator.
18. Synergy-aware corrective controller.
19. Factorial simulation experiments.
20. Robustness/sensitivity analysis.
21. Results figures.
22. Failure-case analysis.
23. Public GitHub repository.
24. Reproducible environment/requirements file.
25. Technical README.
26. Short demonstration video.
27. Final technical report.

25. Experimental Outputs

28. Normal versus impaired reaching trajectories.
29. Joint angles over time.
30. Joint torques over time.
31. Impairment torque versus shoulder-abduction demand.
32. Excess elbow-flexion error versus abduction demand.
33. Controller comparison for each impairment severity.
34. Endpoint error by controller.
35. Robotic effort by controller.
36. Smoothness by controller.
37. Detector estimate versus simulator ground truth.
38. Performance under parameter mismatch.
39. Performance under simulated sensor noise.
40. Torque saturation/constraint violations.
41. Failure-region or operating-envelope plot.

26. Risks and Mitigations

Risk	Impact	Mitigation
Impairment model is physiologically oversimplified	High	Call it phenomenological; validate observable behaviour rather than neural mechanism.
Planar model cannot represent true shoulder abduction	High	Use independent scalar abduction-demand input.

Detector confuses normal coordination with pathological coupling	High	Use task-conditioned residuals and load dependence rather than raw joint correlation.
Controller simply overpowers impairment	High	Measure assistance effort and include generic matched-assistance comparator.
Validation becomes circular	High	Separate verification, internal consistency and external literature correspondence.
Controller becomes unstable	High	Torque/rate limits, damping, timestep testing and robustness analysis.
Results depend on one arbitrary parameter set	High	Parameter sweeps and sensitivity analysis.
Results fail under noise	Medium	Explicit simulated sensor-noise experiments.
AI claims exceed implementation	Medium	Use model-based-control terminology.
Simulation is mistaken for clinical evidence	High	Explicit claims boundary in report and README.
Scope expands into musculoskeletal simulation	High	Preserve reduced-order 2.5D architecture.
ML/RL distracts from core engineering question	Medium	Keep both as future work only.

27. Revised Timeline

Phase	**Focus**	**Target / Exit criterion**
1 - Requirements & mathematical specification	Freeze coordinates, units, equations, u_abd, impairment equation, validation criteria, factors and metrics.	Week 1 - implementation-ready specification.
2 - Arm & voluntary-control verification	Arm dynamics, integration, voluntary reaching, minimum-jerk trajectory, baseline controller.	Week 2 - stable unimpaired reaching and passed verification.
3 - Impairment model	u_abd, alpha, impairment torque, severity/load sweeps; validation.	Week 3 - predefined impairment validation gate passed.
4 - Detector	Implement causal estimator; evaluate against hidden simulator ground truth.	Week 4 - acceptable estimation without privileged state access.
5 - Controllers	Conventional, generic comparator, synergy-aware controller, torque/rate limits.	Week 5 - stable nominal operation.
6 - Experiments & robustness	Severity/load sweeps, directions, ablations, noise and parameter perturbations.	Week 6 - final experiment set completed.
7 - Analysis	Compute metrics; figures; failure-case analysis.	Week 7 - results frozen and interpreted.

8 - Portfolio release	Repository, README, report, reproducibility instructions and video.	Week 8 - release-ready artifact.

28. Definition of Done

- Arm dynamics are implemented and verified.
- Coordinate conventions and units are documented.
- Shoulder-abduction demand is represented independently from planar shoulder angle.
- Voluntary and impairment torques are explicitly separated.
- Impairment parameters are documented.
- Verification tests pass.
- Literature-constrained phenomenological validation is completed.
- Conventional impedance assistance is implemented.
- Generic additional assistance comparator is implemented.
- Synergy estimator is implemented without ground-truth leakage.
- Synergy-aware controller is implemented.
- Torque and torque-rate constraints are active.
- At least three impairment severities and three abduction-demand levels are evaluated.
- Multiple reaching directions are evaluated.
- Noise and parameter-mismatch experiments are completed.
- Task performance, movement quality, robotic effort and detector performance are reported.
- Failure cases are documented.
- Claims remain explicitly restricted to the simulated model.
- Repository can be reproduced from documented instructions.
- Demonstration video and final technical report are complete.

29. Future Development Path

Version	**Direction**
Version 2 - Data-informed model	Use published or accessible experimental data to calibrate the relationship between shoulder-abduction demand and abnormal elbow behaviour.
Version 3 - Patient-specific estimation	Estimate impairment parameters from observed movement rather than assigning severity manually.
Version 4 - Machine-learning estimator	Compare interpretable model-based estimation against learned nonlinear estimators.
Version 5 - Higher-fidelity biomechanics	Introduce 3D shoulder mechanics, gravity, muscle forces and/or OpenSim integration.
Version 6 - Hardware-in-the-loop	Test the control architecture against robotic hardware or an appropriate real-time simulator.
Version 7 - Human research	Subject to collaboration, ethics approval and safety validation, evaluate transfer to real movement data and only then address motor learning, retention, transfer or clinical efficacy.

30. Final Project Position

The project is intentionally not a claim that a simplified simulation can solve post-stroke flexion synergy. Its value lies in asking a precise engineering question within explicit limits:

Can a rehabilitation controller become sensitive not only to whether a reach succeeds, but also to a specified abnormal load-dependent component of how that reach is produced?

The simulation provides a controlled environment in which the impairment is known, its severity can be systematically manipulated, ground truth is available for evaluation, and alternative controllers can be compared under identical conditions.

A successful result would justify further investigation. An unsuccessful result - such as failure to distinguish abnormal from normal coordination, excessive assistance requirements, instability, or no advantage over generic assistance - would also be scientifically informative and will be reported as such.

The purpose of Version 1 is not to demonstrate clinical efficacy. It is to establish whether the modelling, detection and control concept is technically coherent enough to justify the next stage of development.

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